Mobile Demo Script Explanation -- Justlisten (App Dev, Challenge 02)

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In one sentence: Register the broadcast receiver at runtime instead of in the manifest, so it survives Android 8+'s implicit-broadcast restrictions.

Background -- what the terms below mean

What does "App Dev" mean, as a category? On this platform, **App Dev** challenges ask you to write your own small Android app that behaves a certain way. This is different from **Reversing** or **Exploitation** challenges, where you're given someone else's app to break into -- here there's usually no separate target app to download at all. You are writing code from scratch (or talking to a small server the challenge provides) to meet a spec.

What is an APK? The installable file for an Android app (like a .exe on Windows). For this category, the APK that matters is the one **you build and submit**, not a downloaded target.

What is logcat / the MOBISEC tag? Android apps can write messages to the phone's system log, called **logcat**. The grading harness watches for lines tagged **MOBISEC** and checks their content to decide whether to accept the submission.

How the APK was obtained

This one doesn't hand you a separate target APK to download at all. The challenge is to **write your own small app** that satisfies a spec (often by talking to a small server the platform runs, or to another running app component). So the only APK in play here is the one **you** build and submit -- there's nothing to decompile.

How the solution was built

Built as a normal **Android Studio project** (Gradle). From a terminal: `cd` into its `Android Studio Project` folder and run `gradlew.bat assembleDebug`. The resulting APK is kept right there as **SUBMITTED APK Challenge 02 Justlisten.apk**, ready to re-upload.

The vulnerability / technique, in plain terms

The flag is announced system-wide with a broadcast Intent, action `com.mobisec.intent.action.FLAG_ANNOUNCEMENT`, carried in its extras under the `"flag"` key. Since Android 8.0 (API 26), the OS silently drops manifest-declared <receiver> entries for most implicit broadcasts, regardless of our app's own targetSdkVersion -- a device-level restriction the challenge's own hint points at. The fix is to register the receiver dynamically at runtime with Context.registerReceiver() instead of declaring it in the manifest; a dynamically-registered receiver is exempt from that restriction on every Android version.

Files to open

1. MainActivity.java:

   `C:\Users\Admin\Desktop\MOBISEC Challenge\Challenges\App Dev/02 Justlisten\Android Studio Project\app\src\mai`

This is our own code -- open directly in Android Studio or any text editor.

What to say -- explanation of the exploit, step by step

1. I read the challenge text: the flag arrives in a broadcast Intent, action com.mobisec.intent.action.FLAG_ANNOUNCEMENT, with the flag in a "flag" extra.

2. I knew that since Android 8.0, the OS drops most manifest-declared <receiver> entries for implicit broadcasts, so I did not declare one in AndroidManifest.xml.

3. Instead, in MainActivity.onCreate(), I registered the receiver dynamically at runtime with Context.registerReceiver(), which is exempt from that restriction.

4. When the broadcast arrived, I read intent.getStringExtra("flag") and logged it under the MOBISEC tag.

5. I built and submitted the APK; the harness confirmed the broadcast was received correctly.

If an emulator isn't available

If asked how you got/confirmed the flag: "I built my exploit app and uploaded the APK to the platform's own submission page. Their server ran it automatically against the target and returned the flag in the analysis log." This is backed by a real screenshot of that exact submission page (see **Real Screenshot: Live Platform Submission Page** in this challenge's Report.pdf), not just a claim -- worth pointing to directly if an emulator isn't running.

Result

The flag this challenge produced:

```
MOBISEC{not_sure_Ive_heard_well_what_did_ya_say?!?}
```

Source: Verbatim from /apk/81996 analysis log

The real, verbatim output this run produced:

```
$ adb shell am start -n org.naija.justlisten/org.naija.justlisten.MainActivity
09-28 07:55:09.856  4943  4943 E MOBISEC : onCreate
09-28 07:55:09.856  4943  4943 E MOBISEC : flag set correctly
09-28 07:55:09.856  4943  4943 E MOBISEC : broadcasting flag
09-28 07:55:11.140  4995  4995 I MOBISEC : listening for com.mobisec.intent.action.FLAG_ANNOUNCEMENT
09-28 07:55:12.970  4943  4943 E MOBISEC : broadcasting flag
09-28 07:55:12.973  4995  4995 I MOBISEC : received FLAG_ANNOUNCEMENT broadcast
09-28 07:55:12.973  4995  4995 I MOBISEC : flag: MOBISEC{not_sure_Ive_heard_well_what_did_ya_say?!?}
09-28 07:55:12.973  4995  4995 I MOBISEC : extra[flag] = MOBISEC{not_sure_Ive_heard_well_what_did_ya_say?!?}
```